Applied Physics: Photonics Practice Questions

Sample practice set with short answers. This is a practice set, not an official question paper.

Q1. Define population inversion. Why is it needed for laser action?

Answer: Population inversion is the condition in which more atoms are in a higher energy level than in a lower one. Without it, absorption wins over stimulated emission and light cannot be amplified.

Q2. State two properties that make laser light different from ordinary light.

Answer: It is highly monochromatic and highly coherent. It is also highly directional.

Q3. An optical fibre has a core of refractive index 1.50 and cladding of 1.45. Find the numerical aperture and the acceptance angle (launching from air).

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NA = sqrt(n1^2 - n2^2) = sqrt(2.25 - 2.1025) = sqrt(0.1475) = 0.384
acceptance angle = arcsin(NA) = arcsin(0.384) = about 22.6 degrees
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Q4. For the same fibre, find the critical angle at the core-cladding boundary.

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critical angle = arcsin(n2 / n1) = arcsin(1.45 / 1.50) = arcsin(0.9667) = about 75.2 degrees
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Q5. Why does light stay inside an optical fibre?

Answer: Light travels in the core and strikes the core-cladding boundary at an angle larger than the critical angle, so it undergoes total internal reflection again and again along the fibre.

Q6. Name two applications of optical fibres.

Answer: Telecommunication and internet cables, and medical endoscopy.